

ParityFuseNet-MI: Learning Spatio-Temporal Dynamics via Hierarchical Odd-Even Kernel Convolutions for Motor Imagery Decoding

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Abstract—At the forefront of contemporary developments is the development of brain-computer interface (BCI) technology, which has enormous potential for a wide range of applications. The ability to recognize motor imagery (MI) dynamics in electroencephalogram (EEG) stream is a pivotal component of this advancement. This technology is essential to improve human abilities, empower control over devices and surroundings, and help people with disabilities. Although promising, the limited accuracy of brain signal interpretation has prevented BCIs from being widely used in real-world scenarios. In order to classify EEG based MI tasks, this paper presents a novel model that uses odd and even kernel convolution for feature extraction. The independence of this model from artifact removal techniques and minimal preprocessing requirements are two of its main advantages. Using cross-validation with leave-one-subject-out (LOSO), proposed model was evaluated on three standard datasets. BCI contest IV-2a, IV-2b and III-IIIb. The accuracy results were 76. 80%, 78. 34%, and 76. 21%, respectively. These findings demonstrate the strong performance of the model and the promise to further the study of MI decoding.

Index Terms—Electroencephalogram signal classification, Deep learning, Odd-Even kernel convolution, neural signal decoding

I. INTRODUCTION

Motor imagery (MI) refers to mentally simulating movement without actually performing it. It has a significant impact on the creation of brain-computer interface (BCI) systems. These systems are essential for human-machine interaction, assistive technologies, and neurorehabilitation because they enable direct communication between the human brain and external devices [1], [2], [3]. Using EEG signals to accurately classify different MI tasks, such as imagined movements of the bilateral hands, foot, and tongue is one of the main purposes

of BCIs. This capability holds significant promise for aiding individuals with motor disabilities [4]. However, identifying EEG patterns during MI is inherently difficult due to the signals' weak amplitude and their highly variable, non-stationary characteristics, which complicate the process of extracting and analyzing meaningful features [5], [6]. Moreover, EEG recordings are often contaminated by artifacts and external noise, including those caused by muscle activity and eye blinks, which diminish signal clarity [7], [8]. Conventional machine learning techniques frequently fall short in this area, as they depend heavily on manual feature extraction and are often unable to adequately capture the complex temporal and spatial patterns seen in EEG data [9], [10].

A. Contribution

ParityFuseNet-MI, a new deep learning model for MI-EEG signal classification, is presented in this study. Effectively capturing the temporal and spatial fluctuations of EEG data, the suggested method combines multi scale odd and even kernel-based feature extraction inside a convolutional framework. Depthwise convolution and feature concatenation are used in the model to provide effective channel wise feature extraction. Developed for sophisticated BCI systems, this architecture is very effective in identifying intricate spatiotemporal patterns, leading to increased classification accuracy and dependability. As a result, it establishes a new standard for MI-EEG signal analysis performance.

B. Organization

This is how the paper is organised: In Section II, an overview of the relevant work is provided, which also dis-

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cusses earlier approaches and methodologies that are pertinent to this study. Section III provides a detailed description of the model architecture and the procedures that went into creating it. Section IV presents the experimental data, which provide an in-depth analysis of the network's efficiency. Part V covers these results and their ramifications. Section VI, which wraps up the work, outlines the main conclusions and potential directions for further study.

II. PRIOR WORK

BCI devices have the capacity to convert brain signal into orders that may be carried out, opening up revolutionary applications in a variety of domains. This new technology has a wide range of potential applications, from allowing human development to driving assistive technologies and boosting medical advancements. However, the difficult task of accurately categorising EEG signals linked to MI highlights the ongoing need for increased precision and greater cross-individual generalisation. The graph convolutional recurrent attention network presented by Zhang et al. [11] achieved an accuracy of 60.10% when subjected to Leave-One-Subject-Out (LOSO) cross validation (CV). Subject variability was addressed by this design, which successfully extracted spatial temporal information from EEG data using graph-based structures and attention processes. However, its restricted capacity to adjust to new users highlights the continuous difficulty in attaining strong generalisation. The multi branch EEGNet, which combined squeeze and excitation modules, was introduced by Altuwaijri et al. [12] and achieved an accuracy of 69.10%. Classification results were improved by the model's architecture, which was strengthened via branching and attention techniques. To manage the complicated patterns seen in increasingly sophisticated EEG settings, more optimisation is required. Using LOSO CV, Lawhern et al. [13] presented EEGNet, a streamlined model that uses channel-wise and factorized convolutions for effective feature extraction, and it achieved an accuracy of 68.79%. Because it was first tested on a tiny collection of data, questions about its scalability to bigger and more varied datasets still exist, despite its efficacy on small datasets. EEG-TCNet, which was introduced by Ingolfsson et al. [14], reported an accuracy of 69.52%. The model is appropriate for contexts with limited resources because it achieves a realistic balance between computational load and performance. Its performance on larger-scale data is yet unknown, however. The hierarchical Convolutional Neural Network (CNN) based model created by Amin et al. [15] had a lower accuracy of 55.30%, suggesting that there is still opportunity for development in terms of efficiently collecting pertinent EEG characteristics. CNNs and stacked autoencoders were used in a deep learning method by Tabar et al. [16], which achieved a noteworthy accuracy of 75.10% under LOSO validation. Though their model is effective at capturing temporal, spectral, and geographic information, it is not very flexible when applied to other datasets, particularly ones with fewer sample numbers or varied signal properties. Kabir et al. [17] focused on improving MI classification using advanced feature

selection techniques. Their strategy involved multi-subspace randomization and a collaborative unsupervised method (SR-CFS), achieving 69.91% accuracy with LOSO CV. While the method effectively reduced data dimensionality and preserved important features, the narrow dataset scope questions its broader applicability. Despite advancements in multi layer CNN architectures for spatial temporal analysis, when compared to other approaches, the model fared poorly, indicating the need for more creative CNN fusion techniques.

We provide a new approach for feature extraction in a convolutional framework that makes use of a multi-scale odd-even kernel-based network. To facilitate efficient channel-wise feature extraction, our method combines depthwise convolution with feature concatenation. This approach is intended to outperform current models in terms of resilience and classification accuracy, making it a viable option for real-world BCI applications.

III. MATERIALS AND METHODS

The procedures and materials utilised in this study are shown in Figure 1 for a thorough understanding.

A. Data Collection

1) *BCI Contest IV-2a Database (BCI-IV-2a)*: In EEG-based BCI research, the IV-2a data source [18] is a well recognised and often used standard for MI categorisation. It was built by the Graz University of Technology in 2008 and incorporates data for four separate MI tasks: bilateral hands, foot, and tongue motions. With use of 22 EEG electrodes, between 0.5 and 100 Hz, band pass filtering was applied to data gathered from nine individuals at a 250 Hz sampling rate. Every individual took part in two recording sessions, usually one for training and one for testing; there were 288 trials in each session, yielding a dataset of 5184 trials overall.

2) *BCI Contest IV-2b Database (BCI-IV-2b)*: Additionally, Graz University of Technology produced the IV-2b data source in 2008 [19]. The research categorises MI of the bilateral hands. EEG records taken from nine patients was recorded using three channels (C4, C3, and Cz) at a sampling frequency of 250 Hz. Band-pass filtering the EEG activity between 0.5 and 100 Hz was followed by a notch filter at 50 Hz to minimise power line interference. Five recording sessions were completed by each participant, typically including three training sessions and two testing sessions. A total of 600 trials were conducted for each individual, with 120 trials in each session—60 for non-dominant hand and 60 for dominant hand. All nine patients in the dataset have a total of 5400 trials.

3) *BCI Contest III-IIIb Database (BCI-III-IIIb)*: As a component of a prominent lineup for MI driven BCI study, the III-IIIb database [20] focusses on classifying bilateral hands motions using dynamic EEG recordings. Recordings were acquired from three patients using a pair of bipolar EEG leads (C4 and C3) at a sampling frequency of 125 Hz. In order to eliminate power line noise, the signals were filtered utilizing a notch filter, which only allowed frequencies between 0.5 and 30 Hz. There are three recording sessions with four to nine

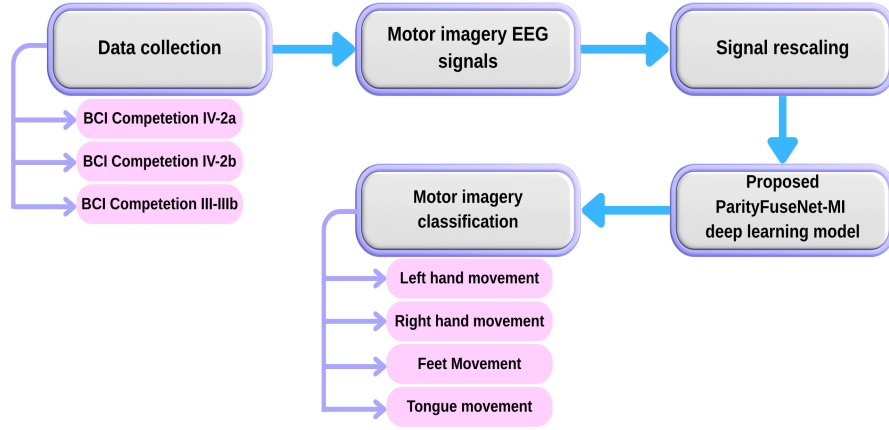


Fig. 1: Pipeline for Motor Imagery EEG Data Processing: From Acquisition to Movement Classification. This process involves data collection from BCI-(IV-2a, IV-2b, and III-IIIb), signal rescaling, and classification of four movement types: bilateral hands, foot, and tongue

runs each for each experiment participant. 160 of the 320 trials completed by subject O3 were assigned to the non-dominant hand, and another 160 to the dominant hand. Both individuals, X11 and S4, performed 1080 attempts, with 540 trials for the non-dominant hand and 540 trials for the dominant hand. A cumulative of 2480 attempts are included in the dataset.

B. Data Preprocessing

To preprocess the IV-2a database, we extract portions out of the General data format (GDF) file, paying attention to the interval from 2.0 to 6.6 seconds within each attempt. This yields in 22 EEG electrodes and 1,150 data samples. Likewise, for each attempt in the IV-2b database, we retrieve data from the time interval from 2.5 and 7.5 seconds, resulting in 1,250 time points and three channels. Lastly, we focus on the BCI-III-IIIb database, which provides 575 time points and two channels, with a temporal frame of 3.5 to 8.1 seconds for each trial. This strategy maintains a steady preprocessing approach customised to the distinctive features of each database. Table I gives a thorough outline of the databases.

Then, employing Min-Max scaling [21], the EEG recording is rescaled to the specified interval of [0, 1] illustrated in equation 1. This ensures that different characteristics or channels have a same scale by standardising the data. It also optimizes model stabilization and minimizes bias caused by dynamic signal scales, making the analysis more objective and implementable.

Consider a sequence $E = \{e_1, e_2, \dots, e_m\}$, where e_j denotes the j -th element of the sequence. The scaled sequence $E^* = \{e_1^*, e_2^*, \dots, e_m^*\}$ is computed as:

$$e_j^* = \frac{e_j - \min(E)}{\max(E) - \min(E)}$$

for all $j = 1, 2, \dots, m$. This scaling procedure is applied to all EEG channels across each trial.

C. Introduced ParityFuseNet-MI Deep Learning Model

We suggest a CNN architecture for classifying EEG motor imagery signals that makes use of a multiscale framework with odd and even kernels as well as depthwise convolution for effective channel-wise feature extraction. In order to extract different feature representations from the input signals, the model uses many convolutional blocks with diverse kernel sizes. Figure 2 shows general design of the suggested model. The EEG signals represented as X_i are numeric tensors of dimensions $C \times T$, where C is the count of EEG channels quantity and T is the entire count of the data points. The suggested model handles these signals. Classifying each input signal X_i into the appropriate class label y_i is the model's main goal.

A labelled dataset S with m EEG trials is used by the model. Every attempt is symbolized by a duo $(X_i, y_i)_{i=0}^m$, where X_i is the input data and y_i is the class label that goes with it. For every given input X_i , the suggested model's job is to forecasts the proper classification label $\hat{y}_i$ from a collection of n potential classes.

1) *Input Layer*: The model accepts input data with shape $(n_{\text{features}}, n_{\text{timesteps}}, 1)$, where n_{features} represents the number of EEG channels, and $n_{\text{timesteps}}$ is the length of the time series data.

$$\text{Input: } X \in \mathbb{R}^{n_{\text{features}} \times n_{\text{timesteps}} \times 1}$$

2) *Convolutional Layers with Even and Odd Kernels*: Several convolutional layers with varying kernel sizes are applied to capture distinct patterns. The first set of convolutional layers uses odd kernel sizes, while the second set uses even kernel sizes:

$$\begin{aligned} X_1 &= \text{Conv2D}(X, \text{filters} = 16, \text{kernel} = (1, 16)) \\ X_2 &= \text{Conv2D}(X, \text{filters} = 16, \text{kernel} = (1, 17)) \\ X_3 &= \text{Conv2D}(X, \text{filters} = 16, \text{kernel} = (1, 18)) \\ X_4 &= \text{Conv2D}(X, \text{filters} = 16, \text{kernel} = (1, 19)) \\ X_5 &= \text{Conv2D}(X, \text{filters} = 16, \text{kernel} = (1, 20)) \end{aligned} \quad (1)$$

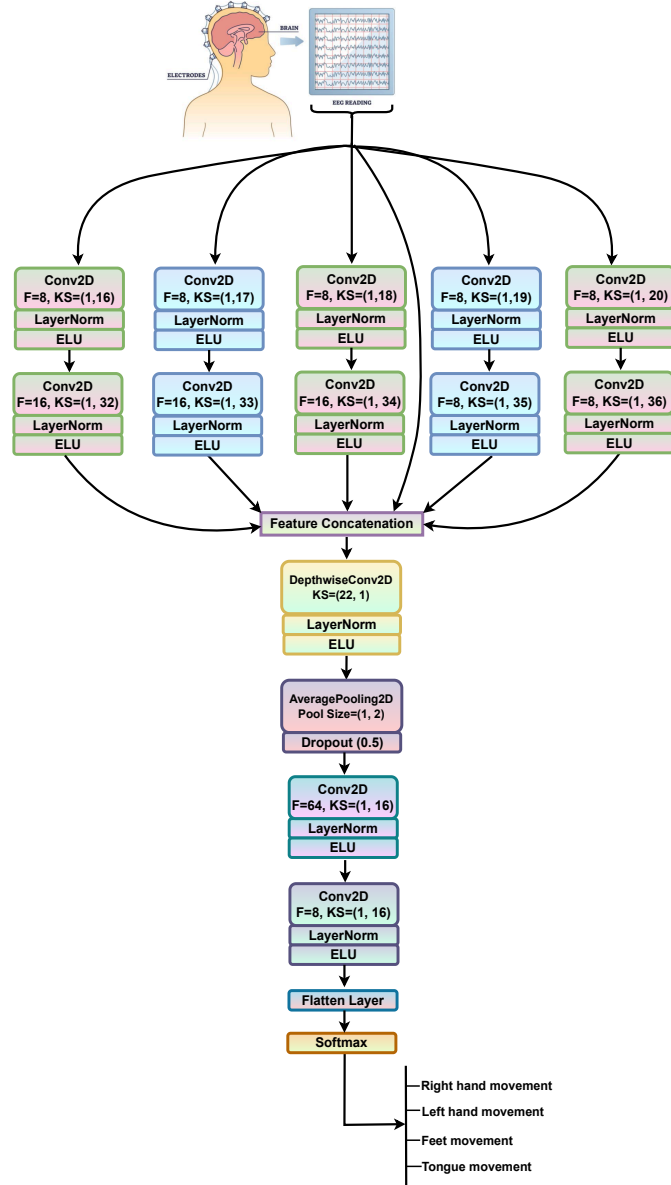


Fig. 2: Proposed ParityFuseNet-MI architecture for MI-EEG decoding. A hierarchical CNN model using odd-even kernel convolutions to capture spatio-temporal EEG features effectively. Designed for classifying bilateral hands, foot, and tongue movements.

TABLE I: A summary of the BCI competition datasets used to classify MI.

Databases	Subjects	Sampling rate (Hz)	Time segment (s)	EEG electrodes	Data points
IV-2a Database	9	250	2-6.6	22	1150
IV-2b Database	9	250	2.5-7.5	3	1250
III-IIIb Database	3	125	3.5-8.1	2	575

Each convolution is followed by Layer Normalization and ELU activation functions to enhance training stability and non-linearity.

3) *Second Convolutional Layer Group*: The outputs from the first group are fed into a second group of convolutional layers with increased filter sizes:

$$\begin{aligned} X_6 &= \text{Conv2D}(X_1, \text{filters} = 32, \text{kernel} = (1, 32)) \\ X_7 &= \text{Conv2D}(X_2, \text{filters} = 32, \text{kernel} = (1, 33)) \\ X_8 &= \text{Conv2D}(X_3, \text{filters} = 32, \text{kernel} = (1, 34)) \\ X_9 &= \text{Conv2D}(X_4, \text{filters} = 32, \text{kernel} = (1, 35)) \\ X_{10} &= \text{Conv2D}(X_5, \text{filters} = 32, \text{kernel} = (1, 36)) \end{aligned} \quad (2)$$

The concatenation of these layers, along with the original input, forms the next layer of the model:

$$DC = \text{Concatenate}(X_6, X_7, X_8, X_9, X_{10}, X) \quad (3)$$

4) *Depthwise Convolution and Pooling*: A depthwise convolutional layer is then applied to extract spatial features more efficiently:

$$X_{dc} = \text{LN}(\text{DConv2D}(DC, \text{kernel} = (n_{\text{features}}, 1))) \quad (4)$$

$$X'_{dc} = \text{AveragePool}(\text{ELU}(X_{dc}), \text{pool} = (1, 2))$$

5) *Final Convolutional Layers*: The final convolutional layers further refine the features before classification:

$$X_{fc} = \text{Conv2D}(X'_{dc}, \text{filters} = 64, \text{kernel} = (1, 16), \text{dilation} = (1, 2)) \quad (5)$$

$$X'_{fc} = \text{Conv2D}(X_{fc}, \text{filters} = 8, \text{kernel} = (1, 16), \text{dilation} = (1, 4))$$

6) *Output Layer*:: Finally, we flatten the output and apply a dense layer with a softmax activation function for classification:

$$\begin{aligned} y_i &= \text{Dense}(\text{Flatten}(X'_{fc}), n_{\text{outputs}}, \text{constraint} = \text{max_norm}(0.25)) \\ \hat{y}_i &= \text{softmax}(y_i) \end{aligned} \quad (6)$$

D. Hyperparameters Settings

We finally found the ideal setup after testing with various kernel sizes and numbers to optimise our model; this is discussed in more depth in the section on the suggested model. Additionally evaluated were various batch sizes, activation functions, loss functions, learning rates, and optimisers. The final setup has a learning rate of 0.001, an Adam optimizer, and ELU activation. 1.52 MB is the size of the model.

E. Evaluation Metrics

Precision, F-1 score, recall, and accuracy are among the metrics that are evaluated in the performance evaluation of the suggested model based on the confusion matrix. The efficacy of the model in classification tasks such as EEG-based MI is shown by these metrics, which are computed using variables like True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN).

$$\begin{aligned} \text{Accuracy} &= \frac{TP + TN}{TP + TN + FP + FN} \\ \text{Precision} &= \frac{TP}{TP + FP} \\ \text{Recall} &= \frac{TP}{TP + FN} \\ \text{F1 - score} &= 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \end{aligned} \quad (7)$$

F. Experimental Setup

To properly evaluate the efficacy of the introduced network, we used LOSO CV over the databases [22]. In each cycle, one subject was excluded for testing, and the model was trained using data from all remaining subjects. This procedure is carefully performed to guarantee that each participant is utilised as the test set exactly once. Therefore, a thorough evaluation of the model's generalizability across many disciplines is conducted.

For the training on Google Colab, a GPU with 12 GB of RAM and the Tensor-Flow framework were used.

IV. EXPERIMENTAL RESULTS

We assessed our proposed network's capability utilizing LOSO CV on three diverse databases. With minimal standard deviations and hovering around 77%, critical measures for the IV-2a, including recall, accuracy, F1-score, and precision, indicate consistent performance among patients. On the IV-2b database, our model also demonstrated strong performance, with accuracy, recall, and precision at about 78%. The accuracy, precision, and recall of our model were about 75%, and its F1-score was 76.80% for the III-IIIb database.

TABLE II: Outcomes of our suggested model utilising LOSO CV on the IV-2a database.

Metric	Mean (%)	Standard deviation
Precision	77.59	3.11
Recall	76.78	2.14
F-1 Score	75.76	3.34
Accuracy	76.80	3.45

TABLE III: Outcomes of our suggested model utilising LOSO CV on the IV-2b database.

Metric	Mean (%)	Standard deviation
Precision	78.60	2.67
Recall	76.67	3.54
F-1 Score	77.23	3.76
Accuracy	78.34	3.54

TABLE IV: Outcomes of our suggested model utilising LOSO CV on the III-IIIb dataset.

Metric	Mean (%)	Standard deviation
Precision	76.33	2.82
Recall	75.22	2.86
F-1 Score	76.80	3.90
Accuracy	76.21	2.83

The model's resilience and adaptability across various datasets are demonstrated by these results, which are detailed in Tables II, III, and IV.

V. DISCUSSION

Due to the intrinsic complexity and variances in EEG signals, MI EEG signal classification poses significant challenges. Our research presents a fresh strategy that successfully

tackles these issues. We used LOSO CV, a reliable method for classifying BCI signals, to validate our model on three different datasets. Across all datasets, the model continuously demonstrated excellent performance.

TABLE V: Comparison of earlier research on the classification of MI.

Approach	Databases	Accuracy (%)
Multi stage CNN [15]	IV-2a	55.30
Attention augmented graph based CNN [11]	IV-2a	60.10
Stacked autoencoder-enhanced CNN [16]	IV-2b	75.10
Multi-path network incorporating squeeze-and-excitation blocks [12]	IV-2a	69.10
SRCFS-LDA MI Classifier [17]	III-IIIb	69.91
Compact CNN [13]	IV-2a	68.79
Temporal CNN [14]	IV-2a	69.52
Proposed ParityFuseNet-MI deep learning model	IV-2a	76.80
	IV-2b	78.34
	III-IIIb	76.21

By comparing our model with current approaches, Table V shows that our method performs more accurately than others. Furthermore, the lightweight nature of the model is beneficial for creating effective BCI technologies, which are essential for implementation in compensatory technology, neurorehabilitation, and communication for people with movement related impairments. Nevertheless, the model's applicability to larger and more varied populations may be constrained by its dependence on a comparatively small dataset. Another significant drawback of the current model is its inability to be interpreted. Future research will concentrate on improving model interpretability and increasing accuracy by integrating bigger datasets.

VI. CONCLUSION

In this paper, a unique odd-even kernel convolution model for EEG-based MI classification is presented. Achieving accuracy rates of 76.80%, 78.34%, and 76.21% on the three diverse databases, correspondingly, the model performs well when using LOSO CV. These results indicate the network's accuracy in classifying MI EEG data. By providing a practical and efficient answer for movement categorisation problems, this method indicates a major progress in MI study. With uses in neurorehabilitation and assistive technology, it has great potential to further BCI technologies. Further study will focus on addressing dataset size and explainability challenges to enhance the network's effectiveness and adaptability.

ACKNOWLEDGEMENT

This research was supported by the Ministry of Science and Technology, Dhaka, Bangladesh, under Grant Number: SRG242277.

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